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MATH 152
Mrs. Bonny Tighe

QUIZ 7
25 points
11.1 – 11.3

NAME Solutions

TA _____ Wed 10/23/2019

YOU MAY NOT USE ANY COMPARISON TESTS ON THIS QUIZ

1. a. State the Geometric Series Test

The geometric series $\sum_{n=1}^{\infty} ar^{n-1}$ is convergent if $|r|<1$ and is divergent if $|r| \geq 1$.

1pt

b. State the test for Divergence

If $\lim_{n\to\infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

1pt

c. State the Integral Test

Suppose f is a continuous, positive, decreasing function on $[1,\infty)$ and let $a_n = f(n)$.
Then (i) If $\int_1^{\infty} f(x)\,dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent
(ii) If $\int_1^{\infty} f(x)\,dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

1pt

d. State the p-series test

The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p>1$ and is divergent if $p \leq 1$.

1pt

2. Find the limits of the sequences and state if the sequence converges or diverges. Is the sequence bounded? If so, what are the bounds. Determine whether the sequence is increasing, decreasing or not monotonic

a) $a_n = \frac{n^3}{3^n}$

$\lim_{n\to\infty} \frac{n^3}{3^n} = 0$

$\frac{1}{3}, \frac{8}{9}, \frac{27}{27}, \frac{64}{81}, \frac{125}{243}$

not monotonic

upper bound $= 1$
lower bound $= 0$ } so Bounded

2pts

b) Find a_n first $\left\{\frac{1}{2}, \frac{3}{5}, \frac{5}{8}, \frac{7}{11}, \ldots\right\}$

$a_n = \frac{2n-1}{3n-1}$

$\lim_{n\to\infty} \frac{2n-1}{3n-1} = 2/3$ so convergent

monotonic increasing

lower bound $= 1/2$
upper bound $= 2/3$ } Bounded

3pts

3. Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2+3k+2}$

$\frac{A}{k+1} + \frac{B}{k+2} = \frac{1}{(k+1)(k+2)}$

$A(k+2) + B(k+1) = 1$

$k = -1 \quad A = 1$

$k = -2 \quad B = -1$

2pts

$\sum_{k=1}^{\infty} \left(\frac{1}{k+1} - \frac{1}{k+2}\right)$

$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}$
$-\frac{1}{3} - \frac{1}{4} - \frac{1}{5} - \frac{1}{6}$

Sum $= 1/2$

I found the sum so

$\sum_{k=1}^{\infty} \frac{1}{k^2+3k+2}$ is convergent

b) $\sum_{k=1}^{\infty} \frac{e^{k+1}}{3^{k+2}}$ $\quad \sum_{k=1}^{\infty} \frac{e\,e^k}{9\,3^k}$

$\sum_{k=1}^{\infty} \frac{e}{9}\left(\frac{e}{3}\right)^k$ is convergent

because $|r| = \frac{e}{3} < 1$

so we can find the sum

$\sum_{k=1}^{\infty} a r^{k-1} = \frac{a}{1-r}$

$\sum_{k=1}^{\infty} \frac{e^2}{27}\left(\frac{e}{3}\right)^{k-1} =$

2pts

$\frac{\frac{e^2}{27}}{1 - e/3} = \frac{e^2}{27-9e}$

8

64 Express $0.\overline{35}$ as a ratio of integers using infinite series.

$.35 + .35\left(\frac{1}{100}\right) + .35\left(\frac{1}{100}\right)^2$ $\quad |r| = \frac{1}{100} < 1$ so convergent and we can find the sum

$$\sum_{n=0}^{\infty} \frac{35}{100}\left(\frac{1}{100}\right)^n$$

$$\frac{\frac{35}{100}}{1-\frac{1}{100}} = \frac{35}{99}$$

2pts

5. Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$$

$f(x) = \frac{e^{1/x}}{x^2}$ is continuous, positive and decreasing for $n \geq 1$

$$\lim_{t\to\infty} \int_1^t \frac{e^{1/x}}{x^2}\,dx \qquad \boxed{u = \frac{1}{x} \quad du = \frac{-1}{x^2}}$$

$$\lim_{t\to\infty} -e^{1/x}\Big]_1^t = \lim_{t\to\infty}\left(-e^{1/t} + e^1\right) =$$

$$-e^0 + e = e - 1$$

3pts

which is a finite limit so

$\int_1^{\infty} \frac{e^{1/x}}{x^2}\,dx$ is convergent

So by the Integral Test

$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$ is also convergent

$$\int_{n+1}^{\infty} f(x)\,dx \le R_n \le \int_{n}^{\infty} f(x)\,dx$$

convergent p-series $p=5>1$

let $f(k) = \frac{1}{k^5}$ which is continuous positive and decreasing for $k \ge 20$.

2pts

6. a) Estimate the error in using the partial sum S_{20} of the series $\sum_{k=1}^{\infty} \frac{1}{k^5}$

$$\lim_{t\to\infty} \int_{20}^{t} k^{-5}\,dk =$$

$$\lim_{t\to\infty} -\frac{1}{4k^4}\Big]_{20}^{t} = \lim_{t\to\infty}\left[-\frac{1}{4t^4} + \frac{1}{4(20)^4}\right] = \frac{1}{4(20)^4}$$

$$\frac{1}{4(21)^4} \le R_{20} \le \frac{1}{4(20)^4}$$

b) What partial sum would be necessary for the above series to have 10^{-4} accuracy?

$$\frac{1}{4n^4} < \frac{1}{10^4}$$

$$\frac{10^4}{4} < n^4$$

$$\sqrt[4]{\frac{10^4}{4}} < \sqrt[4]{n^4}$$

$n > \frac{10}{\sqrt{2}}$ the next higher integer

1pt

7. Fill in the blank: a) If $\sum_{k=1}^{\infty} a_k$ *converges, then* $\lim_{k\to\infty} a_k =$ 0

b) What is the symbol for the 30th partial sum of a series? S_{30}

c) If $\sum_{k=1}^{\infty} a_k$ *diverges and* $\sum_{k=1}^{\infty} b_k$ *converges then* $\sum_{k=1}^{\infty}(ca_k + db_k)$ Diverges

1pt each

Circle True of False

TRUE FALSE d) If $\lim_{k\to\infty} a_k = 0$ *then* $\sum_{k=1}^{\infty} a_k$ converges.

TRUE FALSE e) The Integral Test can be used to test $\sum_{k=2}^{\infty} \frac{k^2 - k^4}{k^6 - 1}$ for convergence

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1. a. State the Geometric Series Test

b. State the test for Divergence

c. State the Integral Test

d. State the p-series test

2. Find the limits of the sequences and state if the sequence converges or diverges. Is the sequence bounded? If so, what are the bounds. Determine whether the sequence is increasing, decreasing or not monotonic

a) $a_n = \frac{3^n}{n^3}$

$\lim_{n\to\infty} \frac{3^n}{n^3} = \infty$ so Divergent

$\frac{3}{1}, \frac{9}{8}, \frac{27}{27}, \frac{81}{64}, \frac{243}{125}$

not monotonic

not bounded

b) Find a_n first $\left\{\frac{1}{2}, \frac{4}{6}, \frac{7}{10}, \frac{10}{14}, \ldots\right\}$

$a_n = \frac{3n-2}{4n-2}$ $\lim_{n\to\infty} \frac{3n-2}{4n-2} = 3/4$

So convergent

monotonic increasing

Bounded - lower bound $1/2$

upper bound $3/4$

3. Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2+4k+3}$

$\frac{A}{k+1} + \frac{B}{k+3} = \frac{1}{(k+1)(k+3)}$

$A(k+3) + B(k+1) = 1$

$k=-1$ $A = 1/2$

$k=-3$ $B = -1/2$

$\sum_{k=1}^{\infty} \frac{1/2}{k+1} - \frac{1/2}{k+3}$

$\frac{1/2}{2} + \frac{1/2}{3} + \frac{1/2}{4} + \frac{1/2}{5} + \frac{1/2}{6}$

$-\frac{1/2}{4} - \frac{1/2}{5} - \frac{1/2}{6}$

$\frac{1}{4} + \frac{1}{6} = 5/12$

I found the sum so

$\sum_{k=1}^{\infty} \frac{1}{k^2+4k+3}$ is Convergent

b) $\sum_{k=1}^{\infty} \frac{e^{k+2}}{3^{k+1}} = \sum_{k=1}^{\infty} \frac{e^2 e^k}{3 \cdot 3^k}$

$\sum_{k=1}^{\infty} \frac{e^3}{9}\left(\frac{e}{3}\right)^{k-1}$

is convergent because

$|r| = e/3 < 1$ so we

can find the sum

$\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}$

Sum $= \frac{e^3/9}{1-e/3}$

$= \frac{e^3}{9-3e}$

64 Express $0.\overline{65}$ as a ratio of integers using infinite series.

$$0.\overline{65} = .65 + .65\left(\frac{1}{100}\right) + .65\left(\frac{1}{100}\right)^2 \ldots$$

$$\sum_{n=0}^{\infty} \frac{65}{100}\left(\frac{1}{100}\right)^n$$ Is a convergent geometric series $|r| = \frac{1}{100} < 1$ so we can find the sum

$$\text{Sum} = \frac{a}{1-r} \qquad \frac{\frac{65}{100}}{1-\frac{1}{100}} = \frac{65}{99}$$

5. Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$$

$f(x) = \frac{e^{1/x}}{x^2}$ is continuous, decreasing and positive for $n \geq 1$

$$\lim_{t\to\infty} \int_1^t \frac{e^{1/x}}{x^2}\,dx = \begin{pmatrix} u = 1/x \\ du = -1/x^2 \end{pmatrix}$$

$$\lim_{t\to\infty} -e^{1/x}\Big|_1^t = \lim_{t\to\infty}\left[-e^{1/t} + e^{1/1}\right] =$$

$$-e^0 + e = e-1$$ which is finite so $\int_1^{\infty} \frac{e^{1/x}}{x^2}\,dx$ is convergent

So By the Integral Test $\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$ is also convergent.

$$\int_{21}^{\infty} f(x)\,dx \le R_{20} \le \int_{20}^{\infty} f(x)\,dx$$

6. a) Estimate the error in using the partial sum S_{20} of the series $\sum_{k=1}^{\infty} \frac{1}{k^4}$

$f(k) = \frac{1}{k^4}$ is cont, dec + pos for $k \ge 1$

$$\lim_{t\to\infty} \int_{20}^{t} k^{-4}\,dk = \lim_{t\to\infty} -\frac{1}{3k^3}\Big|_{20}^{t}$$

$$\lim_{t\to\infty} \left[-\frac{1}{3t^3} + \frac{1}{3(20)^3}\right] = \frac{1}{3(20)^3}$$

$$\boxed{\frac{1}{3(21)^3} \le R_{20} \le \frac{1}{3(20)^3}}$$

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7. Fill in the blank: a) If $\sum_{k=1}^{\infty} a_k$ *converges, then* $\lim_{k\to\infty} a_k =$ 0

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